The Main Categories of Numbers

1. Natural Numbers ($\mathbb{N}$)

Definition:
The natural numbers are the counting numbers.

Standard forms:

- $\mathbb{N} = \{1, 2, 3, 4, \ldots\}$
- Some conventions include 0: $\mathbb{N}_0 = \{0, 1, 2, 3, \ldots\}$

Key features:

- Used for counting and ordering
- No negative numbers
- No fractions or decimals (other than whole numbers)

Examples:
1, 7, 42

2. Whole Numbers

Definition:
The natural numbers together with zero.

Set:

- $\{0, 1, 2, 3, \ldots\}$

Key features:

- Includes zero
- Still excludes negatives and fractions

Examples:
0, 5, 19

3. Integers ($\mathbb{Z}$)

Definition:
All whole numbers, both positive and negative, including zero.

Set:

- $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

Key features:

- Includes negatives
- No fractions or decimals

Examples:
−4, 0, 12

4. Rational Numbers ($\mathbb{Q}$)

Definition:
Numbers that can be expressed as a ratio of two integers, where the denominator is not zero.

Form:

- a/b, where a and b are integers and $b \neq 0$

Key features:

- Includes all integers
- Includes fractions and terminating or repeating decimals

Examples:
1/2, −3/4, 5, 0.25, 0.333…

5. Irrational Numbers

Definition:
Numbers that cannot be expressed as a ratio of two integers.

Key features:

- Decimal expansions are non-terminating and non-repeating
- Cannot be written as fractions

Examples:
$\sqrt{2}$, π, e

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6. Real Numbers ($\mathbb{R}$)

Definition:
All rational and irrational numbers combined.

Set:

- $\mathbb{R}$ = all numbers that can be represented on the number line

Key features:

- Includes integers, fractions, and irrationals
- Covers all continuous quantities

Examples:
-7, $1/3$, $\sqrt{5}$, π

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7. Complex Numbers ($\mathbb{C}$)

Definition:
Numbers that consist of a real part and an imaginary part.

Form:

- $a + bi$, where a and b are real numbers and $i = \sqrt{-1}$

Key features:

- Extends the real numbers
- Used to solve equations like $x^2 + 1 = 0$

Examples:
$3 + 2i$, $-i$, 4

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8. Containment Relationship

Each number system is contained within the next:

$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$

This means every natural number is an integer, every integer is rational, every rational number is real, and every real number is complex.

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9. Quick Reference Table

Category	Includes	Excludes
Natural	Counting numbers	0, negatives, fractions
Whole	Natural numbers + 0	Negatives, fractions
Integers	Positives, negatives, 0	Fractions
Rational	Fractions, repeating/terminating decimals	Non-repeating decimals
Irrational	Non-repeating decimals	Fractions
Real	All numbers on number line	Non-real imaginaries
Complex	Real + imaginary numbers	—